MILITARY SPECIFICATION

SWITCH, ELECTRO-EXPLOSIVE, INSTANT/DELAY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for two types of electro-explosive switches, entitled, Switch, Electro-Explosive, Instant/Delay which is used in the Sprint-Spartan warhead sections.

1.2 Classification. Electro-Explosive Switches covered by this specification shall be of the following types and styles.

Type I - Instant

Style 1 - 4 pole
Style 2 - 6 pole
Style 3 - 8 pole

Type II - Delay

Style 1 - 4 pole
Style 3 - 8 pole

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-T-60 - Tape, Waterproof
PPP-B-00636 - Boxes, Fiber
PPP-C-00843 - Cushioning, Material, Cellulosic

MILITARY

MIL-D-20550 - Diatomaceous Earth (DE)

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

FSC 1135

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

DRAWINGS

PICATINNY ARSENAL

9266791 - Switch, Electro-Explosive, Miniature Instant, C-Bridge, 2-Pin, 4-Pole, 2 Contacts Normally Closed.
EL-9266791 - Equipment List.
9266829 - Switch, Electro-Explosive, Miniature, Instant, C-Bridge, 2-Pin, 8-Pole, 2 Contacts Normally Closed.
EL-9266829 - Equipment List.
9266830 - Switch, Electro-Explosive, Miniature, Instant, C-Bridge, 2-Pin, 8-Pole, 1 Contact Normally Closed.
EL-9266830 - Equipment List.
9266831 - Switch, Electro-Explosive, Miniature, Instant, C-Bridge, 2-Pin, 6-Pole, 2 Contacts Normally Closed.
EL-9266831 - Equipment List.
9266833 - Switch, Electro-Explosive, Miniature, 1.55 Second Delay, C-Bridge, 4-Pin, 8-Pole, 4 Contacts Normally Closed.
EL-9266833 - Equipment List.
9266834 - Switch, Electro-Explosive, Miniature, 0.75 Second Delay, C-Bridge, 2-Pin, 4-Pole, 2 Contacts Normally Closed.
EL-9266834 - Equipment List.
9266835 - Switch, Electro-Explosive, Miniature, 0.42 Second Delay, C-Bridge, 2-Pin, 4-Pole, 2 Contacts Normally Closed.
EL-9266835 - Equipment List.
9266836 - Switch, Electro-Explosive, Miniature, 0.03 Second Delay, C-Bridge, 2-Pin, 4-Pole, 2 Contacts Normally Closed.
EL-9266836 - Equipment List.
9266837 - Switch, Electro-Explosive, Miniature, 0.05 Second Delay, C-Bridge, 2-Pin, 4-Pole, 2 Contacts Normally Closed.
EL-9266837 - Equipment List.

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9271826 - Switch, Electro-Explosive, Miniature, Instant, F-Bridge, 2-Pin, 4-Pole, 2 Contacts Normally Closed.
EL-9271826 - Equipment List.
9271827 - Switch, Electro-Explosive, Miniature, Instant, C-Bridge, 4-Pin, 6-Pole, 4 Contacts Normally Closed.
EL-9271827 - Equipment List.

PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specified procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMC Pamphlet 706-111 - Experimental Statistics.
ASPR 7-104-79 - Armed Services Procurement Regulations.

(Application for copies should be addressed to Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-S.)

Academic Press, New York, 1961 - Quantitative Organic Micro-Analysis.

(Application for copies should be addressed to Academic Press, 111 Fifth Avenue, New York, New York 10003).

ASTM-E-148 - Standard Specification for Apparatus for Microdetermination of Nitrogen by Dumas Method.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing. When a definite

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material is not specified, a material shall be used which will enable the switches to meet the performance requirements of this specification. Materials which are not specified herein or in the detailed specification shall be submitted to the qualifying activity on a formal list of materials for approval. Acceptance or approval of any constituent material shall not be construed as a gaurantee of the acceptance of the finished product.

3.2 Components. The components of the electro-explosive switches shall comply with the requirements of their applicable drawings, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawings and herein.

3.2.1 Bridgewire weld integrity. Bridgewire welds shall withstand a pull force of not less than 10 grams before destruction, when tested as specified in 4.4.1.

3.2.2 Glass seal. The glass seals shall not show any cracks extending from the meniscus area adjacent to the pin to the meniscus area adjacent to the surrounding header. The clouding of glass area adjacent to squib pins caused by the bending stress on the glass is acceptable. The diameter of bubbles or dark spots of pigment concentration in the glass shall be less than one-fourth of one of the following distances, as applicable. The distance between pins or the radial distance between one pin and the surrounding header metal. No metal particles shall be present in the glass. Inspection shall be as specified in 4.4.2.

3.2.3 Squib terminal strength on ignition plug assembly (prior to assembling into switch housing). The terminals of the ignition plug shall not break when subjected to the twisting sequence as specified in 4.4.3. The squib bridge wire shall remain securely connected mechanically and electrically to the terminals. After completion of this test the bridgewire shall comply with 3.7.3.1.

3.2.4 Diatomaceous Earth (DE). DE shall be in accordance with MIL-D-20550, Grade A.

3.2.5 Potassium Dinitrohydroxy Hydrobenzofuroxan or commonly referred to as Potassuim Dinitrobenzofuroxan (KDNBF). KDNBF which meets the No-Fire and All-Fire sensitivity requirements specified herein may be utilized for spotting and base charge blends. It shall have been tested within 90 days prior to start of assembly of the electro-explosive switches.

3.2.5.1 No-Fire Sensitivity. The No-Fire sensitivity of KDNBF powder shall be determined by the Bruceton Up and Down method of sensitivity testing. Fifty (50) units shall be tested. The $\bar{X}$ average (50 percent point) shall be equal to or greater than that indicated in Table I. The $\bar{X}$ average minus four sigma shall be equal to or greater than that indicated in the table. Prior to this test, each bridgewire shall comply with 3.7.3.1. The current shall be applied to the bridgewire

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for a minimum duration of 2 minutes. The No-Fire sensitivity shall be determined at 185 + 5 degrees fahrenheit (°F) when tested as specified in 4.4.4. An ignition plug assembly, Figure 1, shall be used for this test. This test shall be performed on all KDNBF batches of powder manufactured. Results shall be recorded with photographs or other approved recording device.

TABLE I

Bridge Type	$\bar{X}$ Average (amps)	$\bar{X}$ minus 4 Sigma (amps)
C, 2-pin	0.126	0.105
C, 4-pin 2 bridgwires in series	0.126	0.105
F, 2-pin	1.25	0.85

3.2.5.2 All-Fire sensitivity. The all fire sensitivity of the KDNBF powder shall be determined by the Bruceton Up and Down method of sensitivity testing. Fifty (50) units shall be tested. The $\bar{X}$ average (50 percent point) shall be equal to or less than that indicated in Table II. The $\bar{X}$ average plus four sigma shall be equal to or less than that indicated in the table. Prior to this test, each bridgewire shall comply with 3.7.3.1. Pulse shall be square wave shape of a duration of 1.9 to 2.1 milliseconds. The test shall be performed at -65 + 5°F and as specified in 4.4.5. An ignition plug assembly, Figure 1, shall be used for this test. This test shall be performed on all KDNBF batches of powder manufactured. Results shall be recorded with photographs or other approved recording device.

TABLE II

Bridge Type	$\bar{X}$ Average (amps)	$\bar{X}$ plus 4 Sigma (amps)
C, 2-pin	0.780	0.840
C, 4-pin	0.780	0.840
F, 2-pin	5.10	5.50

3.2.6 KDNBF/DE (50/50) blend.

3.2.6.1 Nitrogen content of dried blend (KDNBF/DE). The percent nitrogen content by weight of the dried blend shall be determined as specified in 4.4.6. This requirement shall apply to each batch of powder blend.

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3.2.6.2 Pressure-versus-time. A pressure-versus-time profile shall be obtained for each batch of KDNBF/DE blend. Peak pressure shall be between 1200 and 2400 pounds per square inch (psi) and rise time (0-90 percent of peak pressure) shall be 2.5 milliseconds maximum. Squib assemblies (Figure 2) shall be utilized for this test. Testing shall be as specified in 4.4.7. Results shall be permanently recorded.

3.2.7 Squib assemblies. Before instant and delay squib assemblies are loaded into electro-explosive switches, samples per Table III of each assembly shall be tested to the current and the times specified. Prior to this test each sample selected shall comply with 3.7.3.1 for each bridgewire. Testing shall be as specified in 4.4.8. Different ignition plugs and function times constitute different tests. See Figures 3 and 4. Results shall be recorded.

TABLE III

Squib Assembly Lot Size	Squib Assembly Sample
Up to 50	10
51 - 100	20
101 - 300	30
301 or above	50

No fire sample at ambient temperature. The no fire requirement for the C-bridge is 0.1 (+.00 -.05) ampere for a period of 5 minutes minimum and that of the F-bridge, 1.0 (+.00 -.05) ampere for 5 minutes minimum. Divide samples equally and function one group at high temperature and the other group at low temperature in accordance with Table IV after one hour of conditioning.

3.3 Assembly. The electro-explosive switches shall comply with all requirements specified on the drawings specified in the contract, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.4 First article samples. Requirements for the submission of first article samples by the supplier shall be as specified in the contract. Each unit of product submitted for approval shall be subjected to the criteria prescribed in 4.2. When specified, the supplier shall record variables data that can be verified. Such records shall be forwarded to this activity (see 6.2), ATTN: **DRDAR-QAN-R,** as required.

3.5 Serialization. In lieu of the six-letter serial code associated with implementation of QAI-PA-1, the contractor shall assign a separate serial number to each electro-explosive switch that is to be subjected to functional testing and for which variables data is required. The serial number shall be at least, but not more than, six digits and shall be affixed to

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to the switch in accordance with instructions on the applicable drawing.

3.6 Test code identification number. The contractor shall provide variable reading data feed back on punch cards using the six digit number designated in the applicable paragraph in Section 3. Punch cards shall be prepared and mailed in accordance with QAI-PA-1.

3.7 Characteristics.

3.7.1 Physical characteristics. The electro-explosive switches shall be fabricated and assembled in accordance with the drawings, and other documents listed in 2.1, as applicable.

3.7.2 Performance.

3.7.2.1 Radiographic examination. Prior to function, all switches shall be radiographically inspected for presence, location and condition of component parts, proper assembly of components, proper solder placement, and freedom from foreign material. All switches subjected to functioning (see 3.7.2.3) shall be radiographically inspected to assure that individual switch contacts are free from deformation detrimental to circuit continuity. Testing and inspection shall be as specified in 4.4.9. All x-ray films, photographs and related records shall be furnished the procuring activity.

3.7.2.2 No-fire current. The electro-explosive switch shall not function or be damaged when the bridgewire is subjected to the current and time duration as specified on the applicable drawing. Testing shall be as specified in 4.4.10. After completion of this test, the switch shall comply with 3.7.3.1.

3.7.2.3 All-fire current and switching time. The electro-explosive switch shall function within the times and limits for the temperature as specified on the applicable drawing when the bridgewire is subjected to the specified current. (The functioning time is the time between first application of the firing current to the bridgewire and the last normally open (N.O.) contact closure). There shall be no evidence of damage (see 6.5). Testing shall be as specified in 4.4.11. After functioning, the switch shall comply with 3.7.3.2, 3.7.4.2, 3.7.4.3, and 3.7.2.1. Variables data shall be recorded to the nearest tenth of a millisecond (.0001 sec) for instant, and nearest millisecond (.001 sec) for the delay type only for functionally tested units. Use test code (identification) number 5231--with Table V (see paragraph 3.6).

3.7.3 Electrical conditions.

3.7.3.1 Bridgewire resistance. The bridgewire resistance of the electro-explosive switch shall be as specified on the applicable drawing when tested as specified in 4.4.12. Variables data shall be recorded to the nearest hundredth of an ohm (.01 ohm) only for functionally tested units. Use test code (identification) number 5310 (Short) and 5311 (Long)--with Table V (see 3.6).

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TABLE IV

Squib Assembly Types	Function Time	High Temp	Low Temp	All Fire (AMP DC)
C, 2-pin Instant	2 Milliseconds max	130 ± 5°F	70 ± 5°F	1.00 + 00 -.05
C, 4-pin Instant	2 Milliseconds max	130 ± 5°F	70 ± 5°F	1.00 + 00 -.05
F, 2-pin Instant	3 Milliseconds max	130 ± 5°F	70 ± 5°F	4.50 + 00 -.05
C, 2-pin .030 Second Delay	.0225 - .0375 sec	130 ± 5°F	70 ± 5°F	1.00 + 00 -.05
C, 2-pin .050 Second Delay	.0375 - .0625 sec	130 ± 5°F	70 ± 5°F	1.00 + 00 -.05
C, 2-pin .420 Second Delay	.378 - .426 sec	130 ± 5°F	70 ± 5°F	1.00 + 00 -.05
C, 2-pin .75 Second Delay	.60 - .90 sec	130 ± 5°F	70 ± 5°F	1.00 + 00 -.05
C, 4-pin 1.55 Second Delay	1.32 - 1.78 sec	120 ± 5°F	90 ± 5°F	1.00 + 00 -.05

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3.7.3.2 Contact resistance. The contact resistance of each closed switch contact (see applicable drawing) shall not exceed 0.008 ohm when tested as specified in 4.4.13. Prior to switch functioning, the switch position shall be as specified on the applicable drawing. After functioning, closed contacts shall in each case be the reverse of that shown on the drawing. Variables data shall be recorded. Variable data shall be recorded to the nearest tenth of a milliohm (0.0001 ohm) only for functionally tested units. Use test code (identification) number 532---, where the fourth digit indicates the position number of the electro-explosive switch contact and the last two digits are given in Table V (see 3.6).

TABLE V

Test Condition	Last Two Digits of Test Code Number
Ambient	00
After Temperature Shock	01
During Acceleration	02
After Acceleration	03
After Shock	04
During Sinusoidal Vibration	05
After Sinusoidal Vibration	06
During Random Vibration (with Electro-Explosive Switch functioning)	07
After Electro-Explosive Switch Functioning	08

3.7.4 Insulation resistance.

3.7.4.1 Before firing. The insulation resistance between squib terminals shorted together and case, between each contact terminal and the case, and between each contact terminal and every other mutually isolated terminal shall be greater than 100 megohms when tested as specified in 4.4.14.

3.7.4.2 Squib transient insulation resistance (instant switches only). After firing, the insulation resistance between one squib terminal and the case, and between the two squib terminals shall be greater than 100 ohms when measured within 10 ± 1 millisecond (ms) after firing the switch, and greater than 500 ohms when measured within 500 ± 10 ms after firing the switch. Testing shall be as specified in 4.4.15. At no time shall the resistance between the squib terminals

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be less than the bridgewire resistance before firing as specified on the applicable drawing.

3.7.4.3 After Firing. The insulation resistance between each contact terminal and the case, and between each contact terminal and every other mutually isolated terminal shall be greater than 10 megohms when tested as specified in 4.4.14.

3.8 Environmental conditions.

3.8.1 Acceleration. The electro-explosive switch shall show no evidence of damage or physical deterioration (see 6.5) when subjected to 250 gravity units (g's) of acceleration in both directions of the three mutually perpendicular axes (total 6 directions) for 2 minutes minimum in each direction (total 12 minutes). Contact chatter shall not exceed 0.4 microsecond. Testing shall be as specified in 4.4.16. After completion of test the switch shall comply with 3.7.3.1, 3.7.3.2, and 3.7.4.1. (When fired, N.O. switch contacts shall close and normally closed (N.C.) switch contacts shall open.) See Figure 5 for switch axes.

3.8.2 Sinusiodal vibration. The electro-explosive switch shall withstand the vibration sweep cycling frequency ranges listed in Table VI applied along each of the three mutally perpendicular axes (total 3 directions) without evidence of damage (see 6.5) or impairment of operations. Contact chatter shall not exceed 0.4 microsecond. Testing shall be as specified in 4.4.17. After completion of test the switch shall comply with 3.7.3.1, 3.7.4.1, and 3.7.3.2. (When fired, N.O. switch contacts shall close and N.C. switch contacts shall open.) See Figure 5 for switch axes.

TABLE VI

Frequency (Hz)	G-level
50 - 2000	40 g
2000 - 3000	20 g

Sweep rate - 2 octaves/minute (upward sweep only).

3.8.3 Random vibration (with electro-explosive switch functioning). The electro-explosive switch shall show no damage (see 6.5) when subjected to the vibrations listed in Table VII applied along each of the three mutually perpendicular axes (total 3 directions) (see Figure 5) for 2 minutes minimum in each axis (total 6 minutes). During the frequency range of 50-3000 Hz at 1.0 g^2/Hz level, the switch shall be functioned and comply with 3.7.2.3. Testing shall be as specified in 4.4.18. When fired N.O. switch contacts shall close and N.C. switch contacts shall open. See Figure 5 for switch axes.

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TABLE VII

Frequency (Hz)	G-level
50 - 3000	1.0 g^2/Hz (12 db/octave minimum)
3000 - 4000	at 1.0 g^2/Hz (12 db/octave minimum - control roll off to 4000 Hz)

54.3 g (+15% -5%) root-mean-square (rms) overall.
Filter equalization; 50 Hz to 3000 Hz: ± 3 db

3.8.4 Mechanical shock. The electro-explosive switch shall show no evidence of damage (see 6.5) and N.O. switch contacts shall not close and N.C. switch contacts shall not open when subjected to shocks of 250 g's sawtooth or 1/2 sine shock pulse in both directions of the three mutually perpendicular axes (total 6 shocks) for 6 ± 1 milliseconds in each direction. Contact chatter shall not exceed 0.4 microsecond. Testing shall be as specified in 4.4.19. After completion of test, the switch shall comply with 3.7.3.1, 3.7.4.1, and 3.7.3.2. See Figure 5 for switch axes.

3.8.5 Temperature shock. The electro-explosive switch shall withstand a temperature change from +185 ± 5°F to -65 ± 5°F to +185 ± 5°F without evidence of physical damage (see 6.5) when tested as specified in 4.4.20. After completion of this test the switch shall comply with 3.7.3.1, 3.7.4.1, 3.7.3.2, and 3.2.2.

3.9 Seal integrity.

3.9.1 Fine leakage. The electro-explosive switch shall not exceed leakage rate greater than 1 x 10^{-6} atmospheric cubic centimeters per second (atm cc/sec) when subjected to a differential of 1 atmosphere. Testing shall be as specified in 4.4.21. After completion of test, the switch shall comply with 3.9.2.

3.9.2 Gross leakage. The electro-explosive switch shall show no evidence of leakage when the switch is immersed in 2 inches of water (in lieu of silicone oil) under a reduced atmospheric pressure of 10 ± 2 inches of mercury for not less than one minute. Testing shall be as specified in 4.4.22.

3.10 Standards of manufacture. Each electro-explosive switch lot shall contain only switches fabricated from homogenous lots of material manufactured under the same conditions and with materials produced from the same powder batch and by the same process. Intermixing, whereby one lot contains switches manufactured with powder from a different powder lot, shall be strictly forbidden. For lot control and traceability, material identity shall be maintained throughout manufacture and to the end item (see 6.2).

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3.11 Workmanship.

3.11.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.11.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.11.3 Burrs. Parts shall be free from burrs.

3.11.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.11.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.11.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted.

4.2.1.1 Subassembly components. The contractor shall submit the following subassembly components for first article testing prior to assembling the samples of electro-explosive switches for end item first article testing.

a. Bridgewires. A random selection of ten (10) sample units per lot of one hundred (100) welded plugs shall be selected and tested per 3.2.1. If the bridgewires comply with 3.2.1, the lot represented by the samples shall be considered to be accepted. The remainder of this accepted lot of welded plugs may be used in the production lot.

b. Switch terminal strength on ignition plug assemblies and contact pin holders prior to assembling into switch housings. A randomly selected sample of five (5) ignition plug assemblies and five (5) contact pin holders per lot of one hundred (100) of each style shall meet the requirements of 3.2.3. If the items comply with 3.2.3, the lot represented

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by the samples shall be considered acceptable. The remainder of these accepted lots of ignition plug assemblies and contact pin holders may be used in the production lot.

c. Potassium Dinitrobenzofuroxan (KDNBF). Fifty (50) units of ignition plug assemblies for no-fire test (see 3.2.5.1) and fifty (50) units of ignition plug assemblies for all fire test (see 3.2.5.2) shall be made from the same batch of KDNBF powder and shall be subjected to the Bruceton Up and Down method of sensitivity testing (see AMC Pamphlet specified in 2.2). If the averages specified in tables in 3.2.5.1 and 3.2.5.2 are met the KDNBF powder batch from which the 100 units were made is acceptable and shall be used for end items first article and production lot units. If the averages and sigmas are not met, the KDNBF powder batch so tested shall be rejected.

d. Diatomaceous Earth (DE). Prior to use of DE in the KDNBF/DE blend, objective evidence (see 6.2) shall be provided to assure that the DE meets the requirements of MIL-D-20550, Grade A.

e. KDNBF/DE (50/50) Blend. The batch of KDNBF/DE blend shall comply with the requirements of 3.2.6.1. In addition, a sample of five (5) squib assemblies (see Figure 2) from each batch of powder blend shall be tested to the requirements of 3.2.6.2. If the batch of blend fails to meet the requirements of 3.2.6.1 and 3.2.6.2, the batch shall be rejected. If the batch of blend meets the requirements of 3.2.6.1 and 3.2.6.2, it shall be used for end item first article and production lot units.

f. Squib assemblies. Sample (see 4.3.3.7 and Table X) squib assemblies shall be selected from each lot of squib assemblies (see Figures 3 and 4) made from the parts qualified above and tested to the requirements of 3.2.7. If the samples meet the requirements of 3.2.7, the squib assemblies are qualified for end item first article and production lot units.

4.2.1.2 Electro-explosive switch assemblies. After the above subassembly components are qualified as being suitable for first article/production lot application, the contractor shall assemble and submit six (6) consecutively produced electro-explosive switch assemblies per drawing specified in the contract for first article testing to the testing facility designated in the invitation for bids or request for proposal. The electro-explosive switches shall be fabricated from the components qualified above.

4.2.2 Inspection and testing to be performed. Each first article sample unit and component shall be examined for the defects/characteristics specified in 4.3.2 utilizing inspection methods contained therein and tested per Tables VIIIa and VIIIb as applicable, in the order specified. See paragraph 6.6 for unit identification.

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.

4.2.4 Rejection. If any subassembly or switch fails to comply with any of the applicable requirements, the first article samples shall be rejected.

4.3 Quality conformance inspection (production lot).

4.3.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

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TABLE VIII a

First Article Tests and Inspection of Subassembly Components

Component	Examination or Test	Applicable Requirement	Applicable Test Method	Frequency ofTest
Bridgewire	Weld Integrity	3.2.1	4.4.1	See 4.2.1.1 a
Ignition Plug Assemblies and Contact Pin Holders	Switch Terminal	3.2.3	4.4.3	See 4.2.1.1 b
KDNBF	No-fire sensitivity	3.2.5.1	4.4.4	See 4.2.1.1 c
	All-fire sensitivity	3.2.5.2	4.4.5	See 4.2.1.1 c
DE	Verification of objective evidence (see 6.2)	3.2.4	4.3.2.1	See 4.2.1.1 d
KDNBF/DE Blend	Nitrogen Content	3.2.6.1	4.4.6	See 4.2.1.1 e
	Pressure vs. Time	3.2.6.2	4.4.7	See 4.2.1.1 e
Squib Assemblies	No-fire	3.2.7	4.4.8	See 4.2.1.1 f
	All-fire	3.2.7	4.4.8	See 4.2.1.1 f

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TABLE VIII b

First Article Tests of Electro-Explosive End Item Switches

Examination or Test	Applicable Requirement	Paragraph Test Method	Frequency of Test Sampling	Frequency of Test All Units	Remarks
1. Classification of Defects/ Characteristics	4.3.2.1*	As specified	X		See Table IX
2. Glass seal	3.2.2	4.4.2		X	6 units per dwg
3. Fine leak	3.9.1	4.4.21		X	6 units per dwg
4. Gross leak	3.9.2	4.4.22		X	6 units per dwg
5. No-fire current	3.7.2.2	4.4.10		X	6 units per dwg
6. Bridgewire resistance	3.7.3.1	4.4.12		X	6 units per dwg
7. Contact resistance	3.7.3.2	4.4.13		X	6 units per dwg
8. Insulation resistance (before firing)	3.7.4.1	4.4.14		X	6 units per dwg
9. Radiographic	3.7.2.1	4.4.9		X	6 units per dwg
10. Temperature shock	3.8.5	4.4.20		X	6 units per dwg
11. Acceleration	3.8.1	4.4.16		X	6 units per dwg
12. Sinusoidal vibration	3.8.2	4.4.17		X	6 units per dwg
13. Mechanical shock	3.8.4	4.4.19		X	6 units per dwg
14. Random vibration (with electro-explosive switch functioning)	3.8.3	4.4.18		X	2 units per dwg for each of 3 axes (total 6 units per dwg)

*NOTE: Only lots which have complied with these examinations/requirements shall be used for First Article Testing. The sample units used for the requirement tests shall not be included in First Article or Production Lot acceptance testing.

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4.3.1.1 Inspection lot. Lot, or batch, and its formation, size, and presentation are described in MIL-STD-105. For the purpose of this specification, lot shall mean "Inspection lot" and shall also be the quantity of parts, subassemblies, or assemblies from the same lot (see 3.10).

4.3.2 Product inspection. Critical defects shall require 100 percent inspection of all items in the inspection lot. If any electro-explosive switch assembly fails to comply with the applicable critical defects inspection, it shall be classified defective and removed from the lot. Samples shall be selected at random from each inspection lot in accordance with the applicable major or minor defects Table IX, and inspected for all defects in the applicable classification. Any sample found to be defective shall reject the inspection lot. Sequence for the inspection of defects in each classification need not be followed as long as all defects inspections are performed.

TABLE IX

Major and Minor Defects Inspection

Inspection Lot Size (Includes Samples for Destructive Testing)	Sample Size
1 - 27	All (entire lot)
28 - 33	16
34 - 41	17
42 - 54	18
55 - 76	19
77 - 121	20
122 - 273	21
274 and over	22

FOR MAJOR DEFECTS APPLICATION: No defect in any sample is allowed. Any sample found to be defective shall reject the inspection lot.

FOR MINOR DEFECTS APPLICATION: Two or more sample units having one or more defects shall reject the inspection lot.

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4.3.2.1 Classification of defects. Critical, major, and minor defects are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Diatomaceous Earth (see MIL-D-20550).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Objective evidence (see 6.2) that Diatomaceous Earth complied with 3.2.4	Visual
Minor:	None defined.	

4.3.2.1.2 Switch, Electro-Explosive, Miniature, Instant/Delay (see drawing listed in 2.1, as applicable, and paragraph 6.2).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Height of pins above contact pin holder incorrect	Gage
	102. Pin diameters incorrect	Gage
	103. Location of pins incorrect	Gage
	104. Pins flattened or cracked	Visual
	105. Switch case damaged	Visual
	106. Markings illegible, incorrect or missing	Visual
	107. Objective evidence that approval certificate has been issued that parts and materials used are resistant to the effects of high levels of nuclear radiation (see 6.2)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.11)	Visual
	202. Serialization not in accordance with 3.5	Visual

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4.3.2.1.3 Preparation for delivery (see Section 5).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Squib pins not shunted prior to packaging	Visual
102.	Protective plastic sleeve not installed on switches	Visual
103.	Packaging, packing and marking not as specified	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.11)	Visual

4.3.3 Testing. The contractor shall perform all the tests specified in this paragraph. Tests on electro-explosive switches shall be performed in the order specified in 4.3.3.7 and 4.3.3.8. Tests for subassembly components do not have to be performed in the order specified.

Test	Subassembly Component	Switch Assembly	Test Classification
*a. Bridgewire weld integrity	x		Major
*b. Switch terminal strength	x		Major
*c. KDNBF No-fire sensitivity	x		Major
*d. KDNBF All-fire sensitivity	x		Major
*e. Nitrogen content of dried blend (KDNBF Blend)	x		Major
*f. Pressure vs time (KDNBF Blend)	x		Major
*g. Squib assy No-fire	x		Major
*h. Squib assy All-fire	x		Major
j. Seal integrity		x	Major
k. Insulation resistance		x	Major
l. Bridgewire resistance		x	Major

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Test	Subassembly Component	Switch Assembly	Test Classification
m. Contact resistance		x	Major
n. Radiographic		x	Major
o. Glass seal		x	Major
p. Temperature shock		x	Major
q. Random vibration with All-fire current and switching time		x	Major

Of these tests on electro-explosive switch assemblies, a through h and p and q are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of switches required by the contract. The samples to be subjected to tests a thorugh h and p and q shall be randomly selected from each inspection lot where the number to be tested is given in Table XI.

*NOTE: These sampling paragraphs apply when lots of subassembly units or powder batches made for first article testing are insufficient for fabrication of end item production lots.

4.3.3.1 Bridgewire weld integrity (see 3.2.1). A random selection of ten (10) sample units per lot of 100 welded plugs shall be tested as specified in 4.4.1. In case of dual bridgewires, the same sampling of plugs shall apply except that each bridgewire shall be tested independently. If any bridgewire fails to comply with 3.2.1, the lot shall be rejected. The samples shall be tested to destruction.

4.3.3.2 Switch terminal strength on ignition plug assemblies and contact pin holder (prior to assembly into switch housings) (see 3.2.3). A random selected sample of five (5) ignition plug assemblies and five (5) contact pin holders from one hundred (100) assemblies in the lot shall be selected and tested as specified in 4.4.3. If any sample fails to comply with 3.2.3, the lot shall be rejected.

4.3.3.3 No-fire sensitivity (see 3.2.5.1) and All-fire sensitivity (see 3.2.5.2) on KDNBF. Sample of 50 units shall be selected from each batch of KDNBF and tested as specified in 4.4.4, and additional sample of 50 units shall be selected and tested as specified in 4.4.5. Both tests shall be conducted in accordance with the Bruceton Up and Down method of sensitivity testing (see 2.1). If any sample fails to comply with the applicable requirements of 3.2.5.1 or 3.2.5.2, the batch shall be rejected.

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4.3.3.4 Nitrogen content (see 3.2.6.1) on KDNBF/DE Blend. Each batch of KDNBF/DE blend shall be tested as specified in 4.4.6. If any batch fails to comply with 3.2.6.1, the batch shall be rejected.

4.3.3.5 Pressure-versus-time (see 3.2.6.2) on KDNBF/DE Blend. A sample of five (5) squib assemblies from each batch of KDNBF/DE Blend shall be tested as specified in 4.4.7. If any sample fails to comply with the requirement of 3.2.6.2, the batch shall be rejected.

4.3.3.6 Squib assemblies (see 3.2.7). Before instant and delay squib assemblies are loaded into switches, samples of each will be randomly selected and tested as specified in 4.4.8. Different ignition plugs and function times constitute different tests. Failure to meet all-fire and the no-fire requirements (time, temperature, current) as specified in 3.2.7 shall be cause for rejection of the entire lot. Each lot of assemblies shall be tested for acceptance in accordance with Table X.

TABLE X

Squib Assembly Lot Size	Squib Assembly Sample Size
Up to 50	10
51 to 100	20
101 to 300	30
301 or above	50

4.3.3.7 Seal integrity (see 3.9), insulation resistance (see 3.7.4.1), bridgewire resistance (see 3.7.3.1), contact resistance (see 3.7.3.2), radiographic (see 3.7.2.1), and glass seal (see 3.2.2). Each switch in the lot shall be subjected to these tests. If any switch fails to comply with the applicable requirements when tested per 4.4, the switch shall be rejected and removed from the lot. Additionally, bridgewire and contact resistance variable data shall be recorded only for those units which will be subjected to functional tests.

4.3.3.8 Temperature shock (see 3.8.5), and random vibration (see 3.8.3 with all-fire current (see 3.7.2.3). Sample quantities of switches of each drawing type, shall be randomly selected from each inspection lot per Table XI. The lot samples shall be subjected to temperature shock as specified. The lot samples shall then be function tested during random vibration, utilizing approximately equal quantities of sample units in each of the three axis. If any switch fails to comply with the

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applicable requirement, the lot represented by the sample in which failure occurred shall be rejected. Variables data shall be recorded for each sample as specified in the requirements. See paragraph 6.6 for unit identification.

TABLE XI

Production Lot Acceptance

No. of Acceptable Units Required to be Delivered (Per Drawing Type)	No. of Sample Units to be Destructively Tested (Sample Size)	Inspection Lot Total
5 - 8	14	19 - 22
9 - 12	15	24 - 27
13 - 17	16	29 - 33
18 - 24	17	35 - 41
25 - 36	18	43 - 54
37 - 57	19	56 - 76
58 - 101	20	78 - 121
102 - 252	21	123 - 273
253 and over	22	greater than 274

4.3.4 Inspection equipment. The contractor shall supply all inspection equipment. Designs, operating instructions, and calibration instructions of the inspection equipment requires approval by SARPA-QA-N-I prior to use (see 6.3).

4.4 Test methods. The tolerances specified therin are absolute with no allowance for test equipment inaccuracy.

4.4.1 Bridgewire weld integrity. The test load shall be applied to the bridgewire and perpendicular to the glass seal at a minimum rate of 50 grams per minute to determine compliance with 3.2.1. The pull test shall be performed on equipment with an accuracy of ± one (1) percent of full scale. Results shall be recorded and logged.

4.4.2 Glass seal. All glass seals shall be examined with a 10 power magnification for the requirements specified in 3.2.2.

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4.4.3 Squib terminal strength. Each squib terminal on the sample shall be twisted 30 ± 5 degrees clockwise (cw), 60 ± 5 degrees counterclockwise (ccw), and then 30 ± 5 degrees cw. The twisting force shall be applied within 0.40 inch of the end. After completion of the test, the bridgewire shall be tested in accordance with 4.4.12.

4.4.4 No-fire sensitivity. The ignition plug assembly (Figure 1) shall be inserted in a temperature chamber and stabilized at 185 ± 5°F (see 6.4). While at this temperature, the bridgewire shall be subjected to a continuous direct current (dc) for 2 minutes minimum to determine compliance with 3.2.5.1. All results shall be recorded and logged for bridgewire resistance, current (no fire/fire), part number, powder batch number, and Bruceton analysis.

4.4.5 All-fire sensitivity. The ignition plug assembly (Figure 1) shall be inserted in a temperature chamber and stabilized at -65 ± 5°F (see 6.4). While at this temperature the bridgewire shall be subjected to a square wave constant current dc pulse duration 1.9 to 2.1 milliseconds to determine compliance with 3.2.5.2. All results shall be recorded and logged for bridge resistance, current (fire/no-fire), part number, powder batch number, and Bruceton Analysis.

4.4.6 Nitrogen content of dried blend. The nitrogen blend shall be determined by Microchemical (Dumas) Method, appearing in a standard or reference chemical instrumentation textbook, utilizing equipment specified in ASTM-E-148 which may be modified to existing laboratory conditions. However, the accuracy and precision of the results shall be equal to or better than the Microchemical (Dumas) Method (see 6.7). Results shall be recorded and shall include powder batch number.

4.4.7 Pressure-versus-time. A time-pressure test fixture with a nominal 0.0197 cubic inch volume shall be utilized to determine compliance with 3.2.6.2. All results shall be recorded and logged by part number, lot number, powder batch, high and low peak force and rise time.

4.4.8 Squib assemblies. Instant squib simulator (Figure 3) and delay squib simulator (Figure 4) shall be utilized to determine compliance with 3.2.7. All results shall be recorded and logged for bridgewire resistance (before functioning), part number, lot number, powder batch, temperature, all-fire current, and function time.

4.4.9 Radiographic examination. All radiographic inspections shall be accomplished by using the film technique method. The switch assembly shall be x-rayed in accordance with procedures specified in MIL-STD-202, Method 209 and in two longitudinal positions, respectively, with terminal pins up and with terminal pins on the side. The x-rayed films and related records shall be arranged to maintain individual switch identities. The switch shall comply with 3.7.2.1.

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4.4.10 No-fire current. The squib terminals shall be subjected to a continuous current of 100 (+.00 -.05) milliamperes dc, or 1.00 (+.00 -.05) ampere dc, as applicable, for 5 minutes minimum. Measurement shall be made with an approved current measuring device to determine compliance with 3.7.2.2. The switch shall then be tested as specified in 4.4.12 to determine if it has functioned.

4.4.11 All-fire current. The squib terminals shall be subjected to a continuous current of 1.00 (+.00 -.05) ampere dc or 4.50 (+.00 -.10) amperes dc, as applicable, to initiate the squib. The functioning time shall commence with the first application of firing current to the squib terminals and shall end with closing of the last normally open circuit. To determine compliance with requirements of 3.7.2.3, the switch functioning time shall be measured with an approved electronic counter. Following this test, the switch shall be tested as specified in 4.4.13, 4.4.14, 4.4.15 (instant only) and 4.4.9, in that order.

4.4.12 Bridgewire resistance. The electrical resistance of each squib bridgewire shall be tested with an approved tester and in accordance with MIL-STD-202, Method 303, with the current limited to 5 ma maximum for 5 seconds maximum.

4.4.13 Contact resistance. The contact resistance of the closed switches shall be measured with an approved tester and in accordance with MIL-STD-202, Method 307, using the four wire method with a test current of 1 ± .05 ampere and open circuit voltage of 10 volts dc maximum.

4.4.14 Insulation resistance before and after firing. The insulation resistance shall be measured with an approved tester and in accordance with MIL-STD-202, Method 302, Test Condition B for 10 seconds minimum or until the meter stabilizes. The test current shall be limited to 5 ma.

4.4.15 Squib transient insulation resistance (instant switches only). The test and firing voltage shall be 28 ± 0.5 volts dc to determine compliance with 3.7.4.2. The insulation resistance shall be photographed or recorded utilizing an approved recording device capable of providing a permanent record.

4.4.16 Acceleration. The switch shall be placed in a suitable fixture secured to an approved centrifuge and subjected to acceleration and times in both directions of the three mutually perpendicular axes in accordance with requirements in 3.8.1. During this test, the switch contacts shall be monitored for contact chatter in excess of 0.4 microsecond with a maximum of 6 volts dc and 10 ma contact current applied. After completion of this test, the switch shall be visually examined for physical damage (see 6.5) and tested in accordance with 4.4.12, 4.4.13, and 4.4.14.

4.4.17 Sinusoidal vibration. The switch shall be placed in a suitable fixture and rigidly mounted to a vibration

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table such as to provide excitation along each of the three mutually perpendicualr axes, and sweep cycled in accordance with requirements of 3.8.2. During this test, the switch contacts shall be monitored for contact chatter in excess of 0.4 microsecond with a maximum of 6 volts dc and 10 ma contact current applied. After completion of this test, the switch shall be visually examined for physical damage (see 6.5) and tested in accordance with 4.4.12, 4.4.13, and 4.4.14.

4.4.18 Random vibration (with switch functioning). The switch shall be placed in a suitable fixture rigidly mounted to a vibration machine and subjected to sweep cycling along each of three mutually perpendicular axes in accordance with the requirements of 3.8.3. During this test, the switch shall be function tested in accordance with 4.4.11. After completion of this test, the switch shall be visually examined for physical damage (see 6.5).

4.4.19 Mechanical shock. The switch shall be placed in a suitable fixture rigidly mounted on a shock machine and subjected to terminal-peak saw tooth shock along each direction of the three mutually perpendicular axes (6 shocks) in accordance with requirements of 3.8.4. During this test, the contacts shall be monitored for contact chatter in excess of 0.4 microsecond with a maximum of 6 volts dc and 10 ma contact current applied. After completion of this test, the switch shall be visually examined for physical damage (see 6.5) and tested in accordance with 4.4.12, 4.4.13, and 4.4.14.

4.4.20 Temperature shock. The switch shall be placed in an approved temperature chamber at 185 ± 5°F and maintained for 4 hours minimum. The switch shall be removed from the high temperature chamber and within 5 minutes placed in a chamber with an integral temperature of -65 ± 5°F and maintained for 4 hours minimum. Within 5 minutes the switch shall be returned to 185 ± 5°F in the high temperature chamber. This sequence constitutes one cycle. Three cycles shall be executed. At the end of the three cycles, the switch shall be returned to ambient temperature and after stabilization (see 6.4), the switch shall be visually examined for physical damage (see 6.5) and tested in accordance with 4.4.12, 4.4.13, 4.4.14, and 4.4.2.

4.4.21 Fine leakage. The electro-explosive switch shall bu subjected to the fine leak test in accordance with MIL-STD-202, Method 112A, Test Condition C, Procedure III b. Transfer of switch from test chamber to radioactivity counter shall be accomplished within 30 minutes maximum. Readings shall be observed to determine compliance with 3.9.1.

4.4.22 Gross leakage. The electro-explosive switch shall be exposed to normal atmosphere and immersed in 2 inches of distilled water. The air over the surface of the water shall be evacuated to a pressure of 10 ± 2 inches of mercury.

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Trapped air pockets may be removed by agitation or jarring during evacuation. While at this pressure for 60 seconds minimum, the switch shall be observed for leakage. Any bubbles coming from within the case shall be considered a gross leak.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level C (see 6.1).

5.1.1 Unit packaging. Prior to intermediate packaging, the squib terminals shall be shunted by the placement of a conductive corrosion resistant spring over the squib terminals on the end of the switch. The terminals shall be inserted between the coils of the spring so the length dimension of the spring runs parallel with the length of the switch. The switch with the spring shall then be inserted into an inert, semirigid, moisture resistant, protective sleeve. The switch and the spring shall be completely inside the sleeve.

5.1.2 Intermediate packaging. The switches shall be packaged in accordance with instructions contained in MIL-STD-1169, however, only switches with the same part number shall be in any one intermediate package.

5.2 Packing. Packing shall be Level C (see 6.1). The packaged items (see 5.1.2) shall be packed in accordance with instructions contained in MIL-STD-1169 except that for fiberboard boxes Grade V3C, Type CF, "B" Flute, Class Weather Resistant, Style RSC of PPP-B-00636 applied. Cushioning material shall conform to PPP-C-00843, Type II, Class A or B, and waterproof tape, 3 inches wide shall conform to PPP-T-60.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169 except that for both intermediate packaging and exterior packing, the marking shall be as follows:

a. Item
b. Quantity
c. Order Number
d. Contract Number
e. Manufacturer/Contractor
f. Lot Number
g. Data packed

In addition to the above markings, the exterior packing only shall have the following:

a. AMC Hazard Class: Unclassified
b. AMC Compatibility Group: General Supply Item
c. DOT Bill of Lading Class: Electro-explosive Switch
d. DOT Container Marking: Electro-explosive Switch-Do Not Ship Parcel Post

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).
c. Safety precautions for ammunition and explosives in accordance with Armed Services Procurement Regulation 7-104.79.

6.1.2 Contract data requirements (see 3.6). Data specified in 3.6 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. The contractor shall furnish objective evidence that the materials used to fabricate any part complies with the specification listed on the applicable drawing. Records of material certifications, contractor quality control, and inspection are to be available for required verifications. Picatinny Arsenal approval that parts and materials used are nuclear resistant shall be certified.

6.3 Inspection equipment. The contractor shall obtain approval of designs, operating instructions, and calibration instructions from the Commander, **US Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.**

6.4 Temperature stabilization. The temperature of the chamber is considered "stabilized" when three consecutive temperature readings, taken at 5 minute intervals, are within ± 4°F of each other.

6.5 Physical damage. Physical damage shall include cracks in glass seal, bent pins, cracks in pins, cracks in body, distortion of body.

6.6 Destructive testing. Each sample unit subjected to destructive testing shall be identified for first article and lot acceptance traceability as delineated in QAI-PA-1.

6.7 Micro-Dumas Method. An acceptable procedure is described by A. Steyemark, Quantitative Organic Micro-Analysis, Academic Press, New York, 1961.

6.8 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2315A dated 12 February 1975.

Custodian:
Army - AR

Preparing activity:
Army - AR

Project No. 1135-A229

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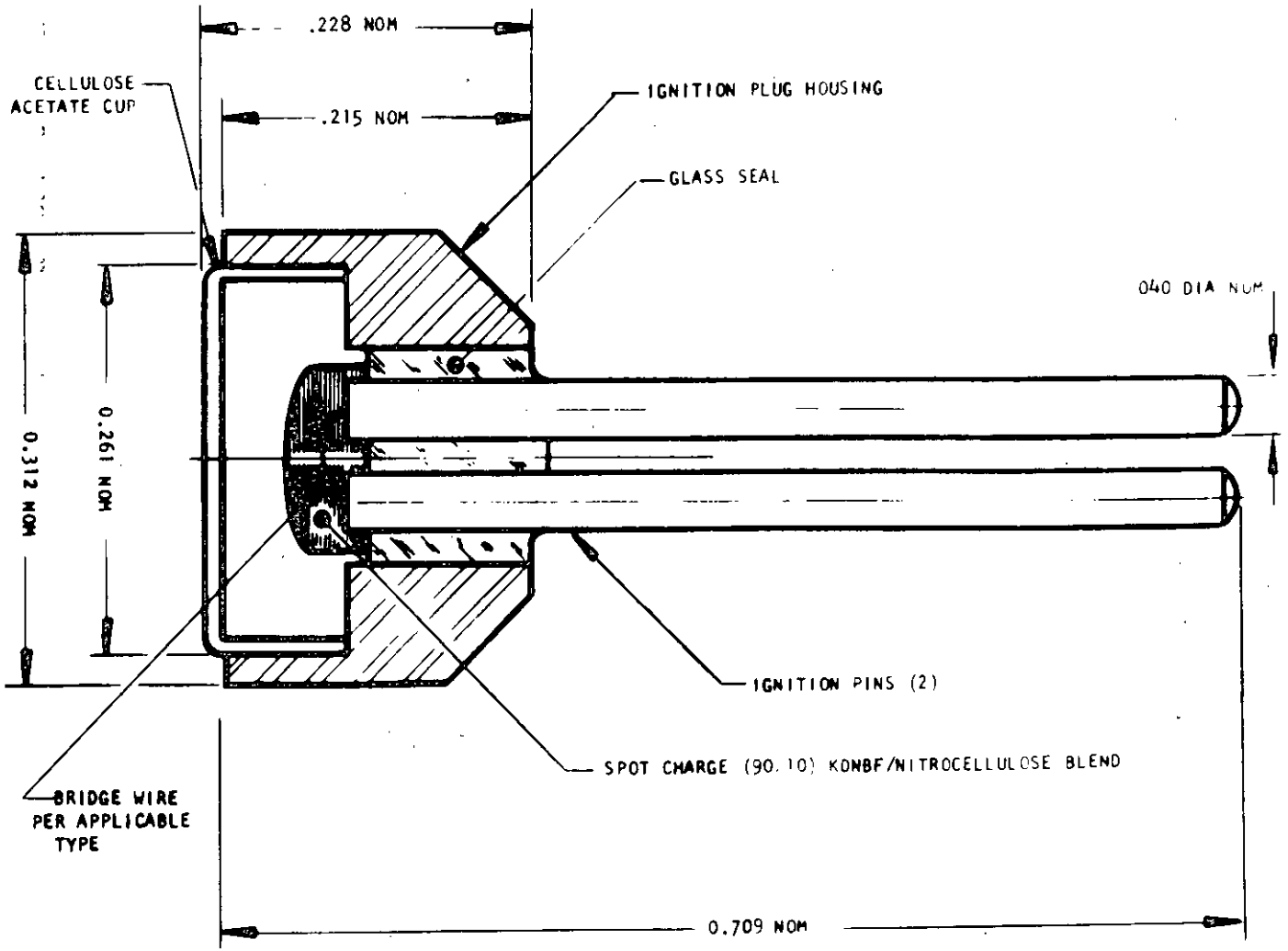


Figure 1. Ignition Plug Assembly

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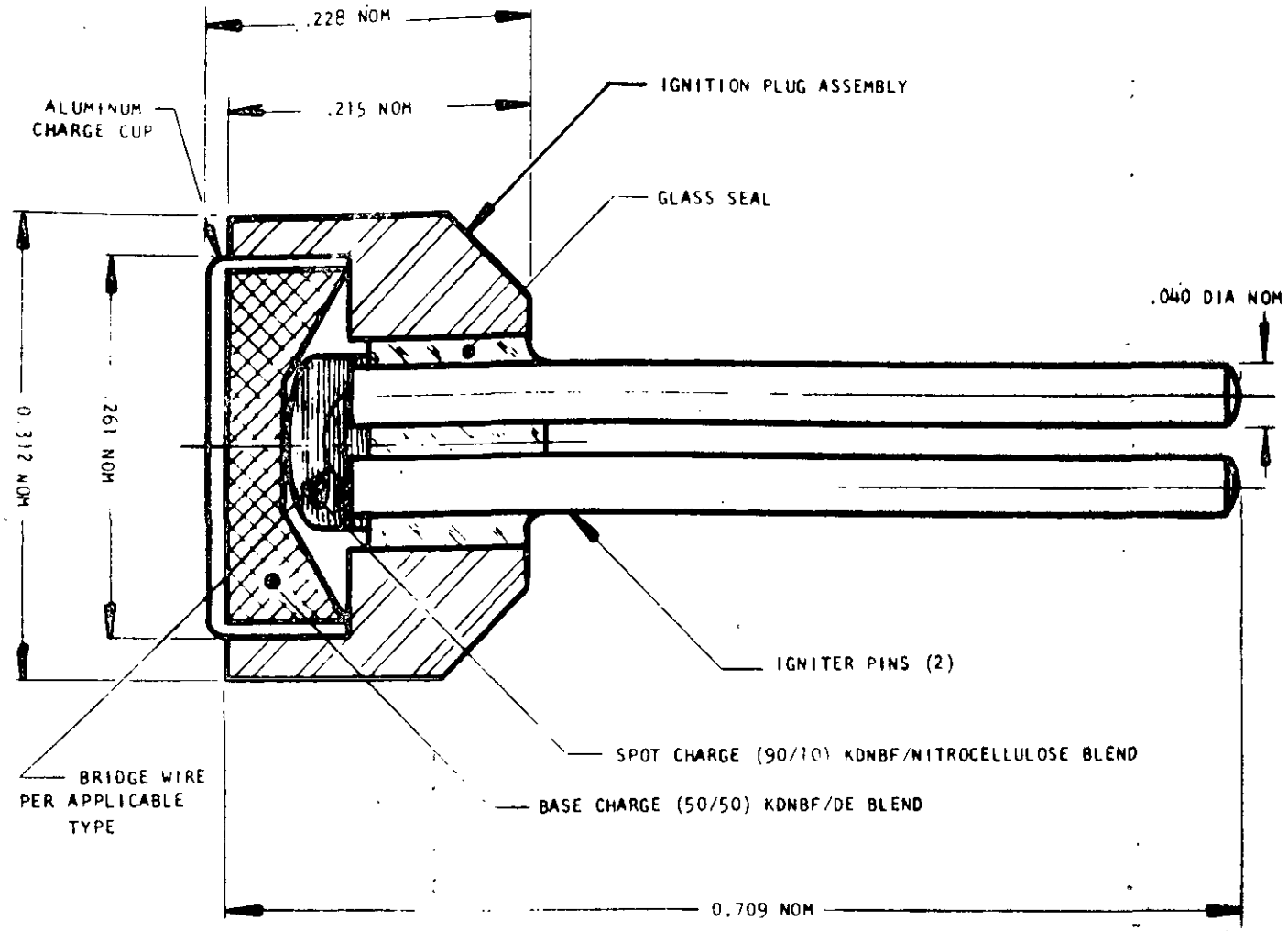


Figure 2. Squib Assembly KDNBF/DE

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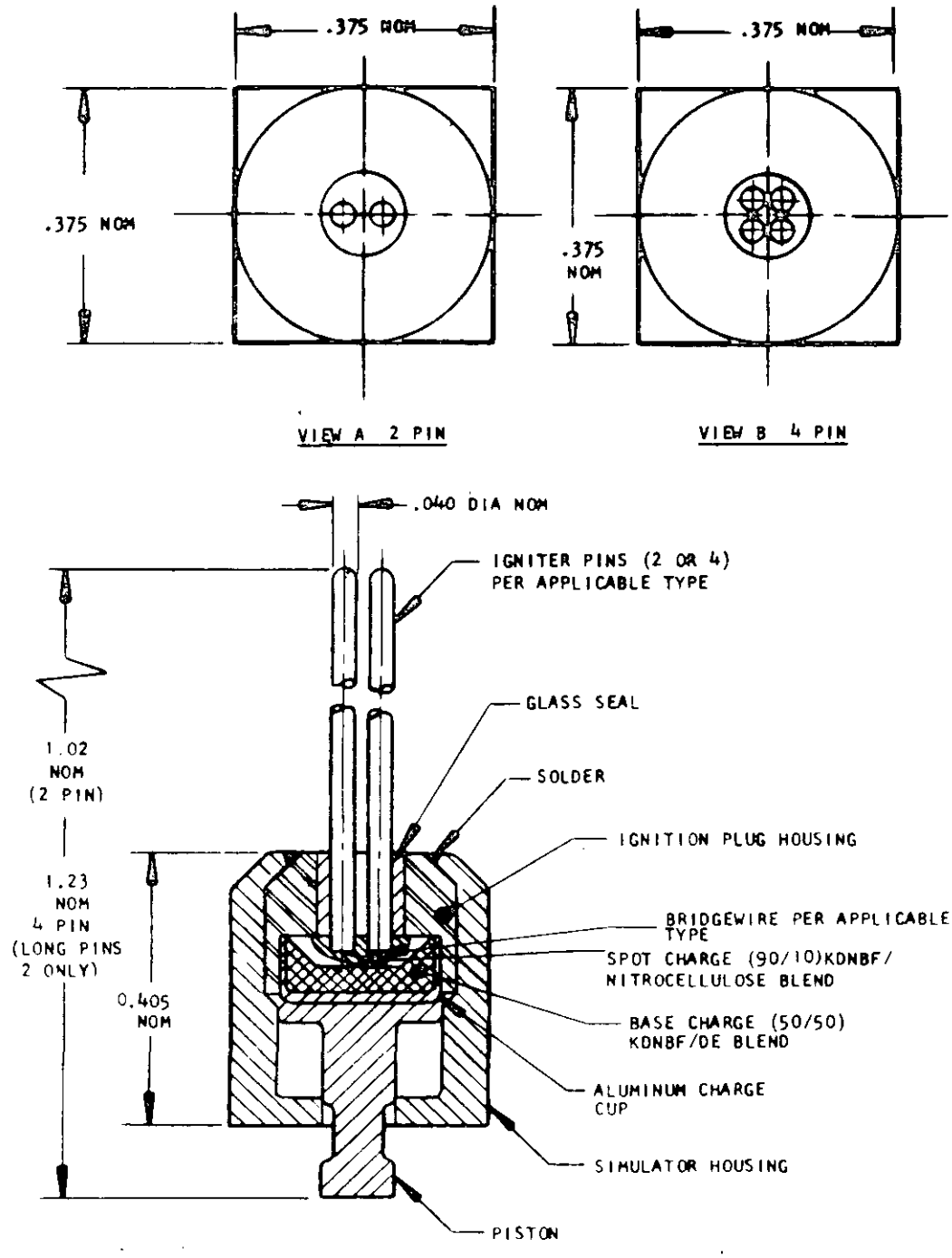


Figure 3. Instant Squib Simulator

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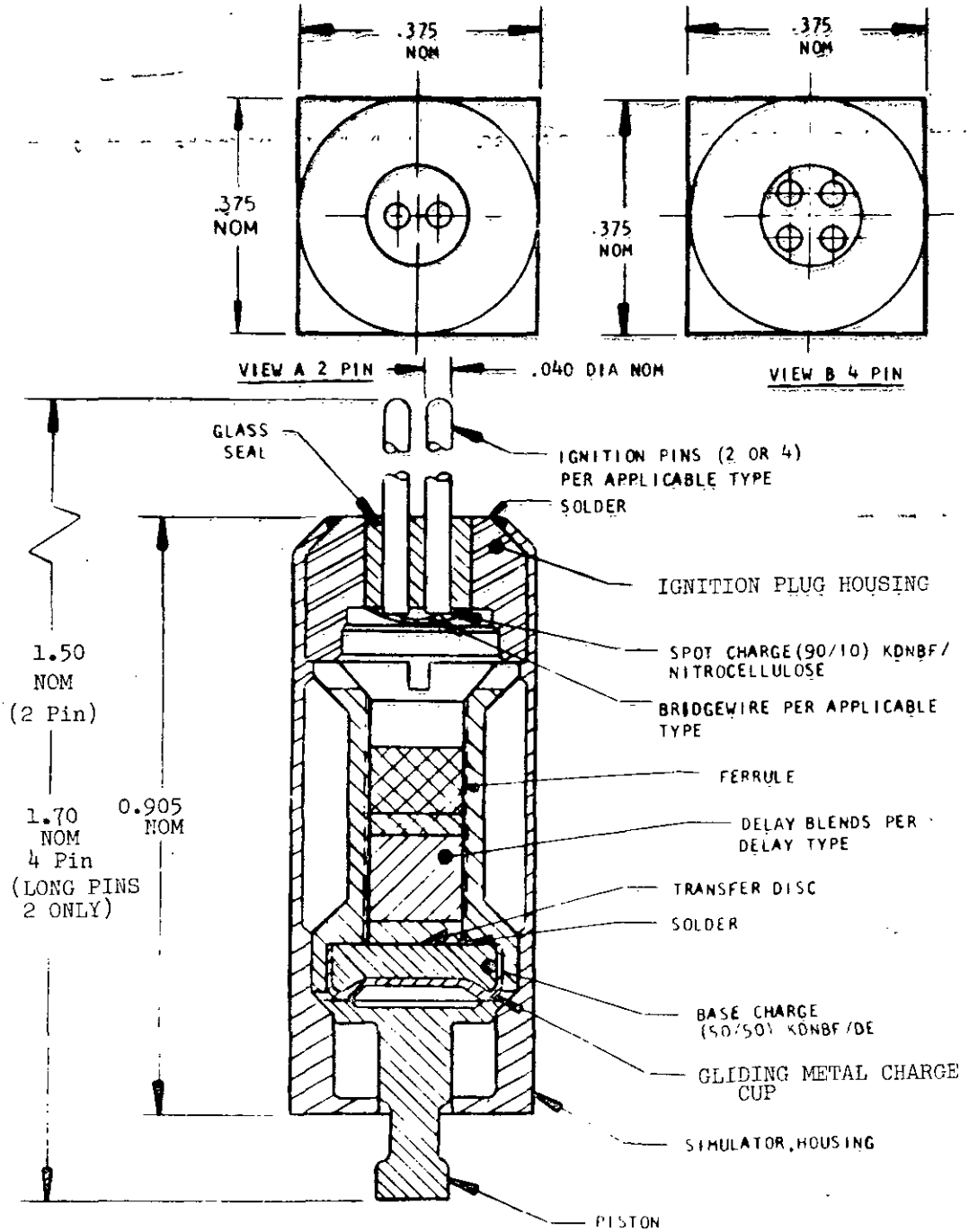


Figure 4. Delay Squib Simulator

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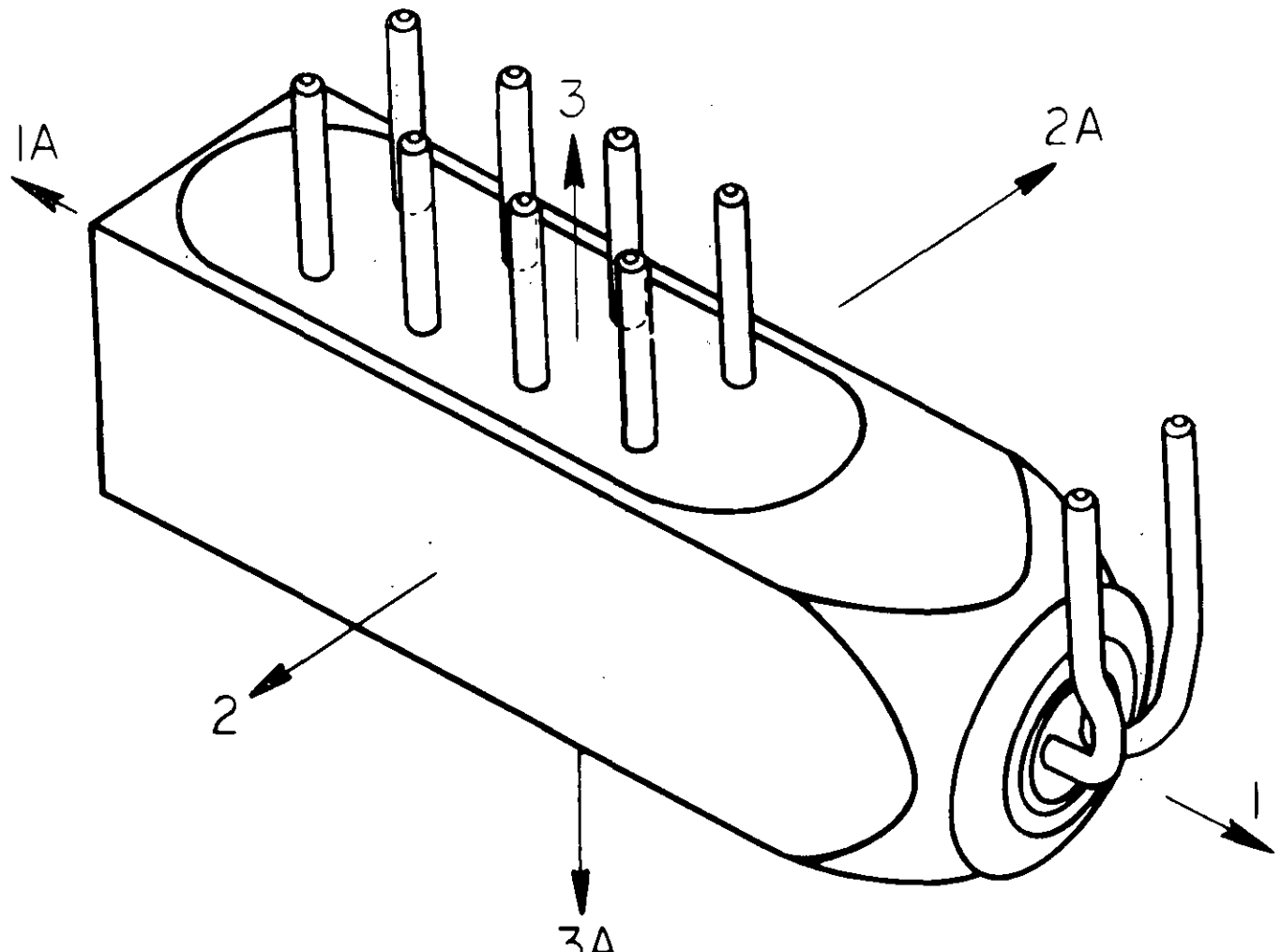


Figure 5. Reference Directions And Axes For Dynamic Tests

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